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Shuffling Disorder
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The problem

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- Shuffling gives card games pseudo-randomness
- Riffle shuffle is the main way
- How can the "pseudo" part be exploited?

The Setup

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- Set of Orders:
$$S = \{[x_1, x_2, \dots, x_n] : \{x_1, \dots, x_n\} = \{1, \dots, n\}\}$$
- Can think of deterministic shuffles as the permutation group S_n
- Shuffle: The $n! \times n!$ transition matrix of a Markov chain on S (Equivalently S_n)
- Riffle step 1: non-deterministically cut the deck into two halves $[x_1, x_2, \dots, x_m]$, and $[x_{m+1}, x_{m+2}, \dots, x_n]$
- Riffle step 2: non-deterministically riffle the two halves onto the table

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Literature Review

" $\frac{3}{2} \log n$ shuffles is enough"

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- Result: " $\frac{3}{2} \log n$ shuffles is enough"
- Calculates exact transition matrix P using GSR model
- After $\frac{3}{2} \log n + \theta$ shuffles, $TV = 1 - 2\Phi(\frac{-2^{-\theta}}{4\sqrt{3}}) + \mathcal{O}(n^{-1/4})$
- This implies a qualitative change in shuffle after $\frac{3}{2} \log n$ as $n \rightarrow \infty$

Critique

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- GSR model makes assumptions and possibly has flaws
- Result applies to n large
- Result is far removed from real life
- To get such a precise result trades off practicality

"7 shuffles is enough"

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- Not much original literature I could find for this
- First time $TV < 0.5$
- "Passes most tests we throw at it" From a numberphile video

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- Seems to be weakly backed up for such a clear result
- Again uses TV, which is arbitrary

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Measures of Disorder of Shuffle

Total Variation

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- $\|P^k - P^\infty\|_{TV} := \frac{1}{2} \max_i \sum_{j=1}^{n!} |(P^k - P^\infty)_{ij}|$
- Measures "distance to the perfect shuffle"
- "rate at which perfectly competent gambler can make money betting on orders the deck will be in after a shuffle"

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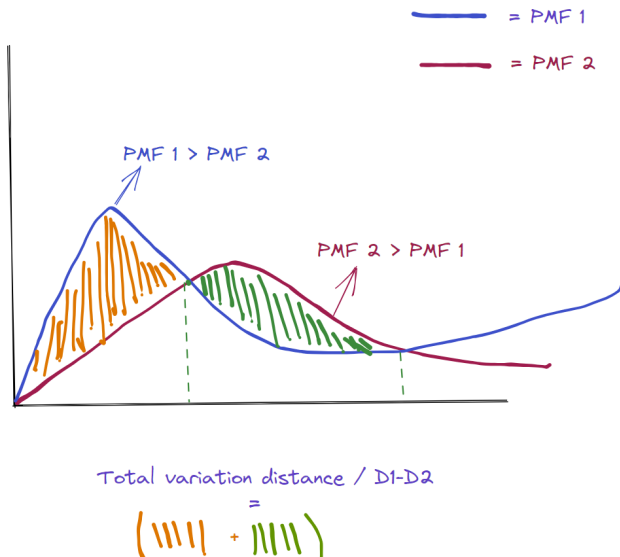


Figure 1: A visualization of the total variation distance.

Information Entropy

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- $U_i^{\text{shuffle}} := \sum_{j=1}^{n!} -p_{ij} \log p_{ij}$
- Measures "how varied the shuffle is"
- "Long term rate at which encoded information can be sent"

Flaws

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- Both require knowing ALL probabilities exactly
- This requires making assumptions
- Treats all permutations equally
- However, they DO take into account how varied the shuffle is

Measures of Disorder of Permutation

Information Entropy 1

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- $U_1 := \sum_{i=1}^{n-1} -y_i \log y_i$
- Measures "how varied the positive distances between adjacent cards are"
- Pros: Fine granularity, seems to agree with humans from testing
- Cons: Doesn't recognise reverses or cycles (cuts!)

Information Entropy 2

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- $U_2 := \sum_{i=1}^{n-1} -y_i \log y_i$
- Measures "how varied the absolute distances between adjacent cards are"
- Pros: Fine granularity, seems to agree with humans from testing
- Cons: Doesn't recognise reverses or cycles (cuts!), less granularity than U_1

$t :=$ Number of transpositions needed to get to original order

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- No "nice" formula
- Measures "how long it takes to put the deck back" in one way
- Pros: Recognises cycles and reversals
- Cons: Distribution of this is very clustered around certain numbers, registers too high for a simple cycle.

$c :=$ Number of Disjoint Cycles

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- $n - t - l$, where $t =$ Number of transpositions needed to get to original order and l is the number of elements fixed by a permutation
- Measures "how many cycles the permutation is made of"
- Pros: Cycles are easy to spot for humans
- Cons: Distribution of this is very clustered around certain numbers, a bit arbitrary

$t' :=$ Number of transpositions needed to get to cycle of original order

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- Again, no "nice" formula
- Measures "how long it takes to put the deck back enough"
- Pros: No longer registers max. disorder from moving top card to bottom
- Cons: Distribution of this is very clustered around certain numbers, more arbitrary than before

Spearman's rank correlation coefficient

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- $r := 1 - \frac{6 \sum_{i=1}^n (X_i - Y_i)^2}{n(n^2 - 1)} = 1 - \frac{6 \sum_{i=1}^n (i - x_i)^2}{n(n^2 - 1)}$
- Measures "how close the cards are to preserving the ascending property"
- Pros: Well studied, seems to mostly agree with human testing, can class the disorder as reversal or not
- Cons: Seems to vary a lot for cycles depending on where the deck is cycled from. Seems to be low (close to 0) for ordered decks but not so consistently close to 0 for "disordered" decks

Number of rising sequences

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- No "nice" formula"
- Measures "how easily and reliably the shuffle can be deconstructed"
- Pros: Linked to riffle shuffles in particular.
- Cons: Not so intuitive at first, doesn't measure our natural idea of "disorder"

Testing the Measures

Conclusion

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- t , c , t' all do not have enough granularity to be useful.
- U_1 and U_2 are heavily correlated, both are useful but U_1 has better granularity.
- r is useful, and is not correlated with U_1
- r depends on distance from starting point, whereas U_1 depends on distance from cards it started near
- rising sequences are not quite what we want, but help pick out the riffle shuffle in particular

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The cut

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- Want a peak around 26, suggesting normal or binomial
- Binomial has evidence for it [5]
- Normal has better flexibility over parameters

The riffle

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- More complex, proportional seems reasonable
- Would be interesting to research more - especially based on average packet size

Experimental Data

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- χ^2 test seems to be flawed
- Dr Beaume's data was slightly flatter, despite this all data fit reasonably
- Gisela's cut of 9 affected her test statistic significantly
- Choice not so important by this measure
- Both are worth analysing

Binomial	Test statistic	p value	95th percentile
Me	18.69345261	0.99999	69.83216
Dr. Beaume	64.622973	0.11238	69.83216
Gisela	4929.160071	0	69.83216

Table: A table of our χ^2 statistics, p values and 5th percentiles for the binomial distribution

Normal	Test statistic	p value	5th percentile
Me	8.205837744	1	67.50481
Dr. Beaume	51.11218098	0.42975	67.50481
Gisela	205.1687494	0	67.50481

Table: A table of our χ^2 statistics, p values and 5th percentiles for the normal distribution

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Deciding on our cutoff point

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- Want a more practical definition of "shuffled enough times"
- Decided to use Aspin-Welch test
- Unlike χ^2 , distributions are normal enough to get good results
- "Given a sample size of 1000, could a statistician tell that the decks weren't randomly generated by looking at their disorder?"
- Need to try this a few times to make sure results are consistent

Binomial

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- Tests were repeated multiple times and results were reproducible
- Using U_1 , there is a big jump in p value after 6 shuffles
- Using r , big jump after k shuffles, but not a consistent increase
- Possibly because riffle shuffles shuffle in a certain "direction", or because r seemed to not measure consistently high for disordered decks
- Using s , Need 11 shuffles to pass the test, and p values generally do increase

k	mean of random samples	mean of samples from k shuffles	x	p
1	4.859977889	2.510088191	260.4443088	0
2	4.859977889	3.915322232	114.184855	0
3	4.859977889	4.564827427	46.84966382	0
4	4.859977889	4.78431633	13.2501361	0
5	4.859977889	4.833408844	5.037899149	2.57E-07
6	4.859977889	4.853949508	1.172483688	0.120571367
7	4.859977889	4.856755739	0.624832806	0.266076092
8	4.859977889	4.857506196	0.469151315	0.31950644
9	4.859977889	4.856412684	0.67155487	0.250972615
10	4.859977889	4.858556473	0.267028054	0.394737655

Table: A table of k , the estimated means of U_1 , test statistics and p values for the test on $k = 1, \dots, 10$ riffles.

Dependence on n

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- It seems intuitive that larger decks take more shuffles, by looking at rising sequences
- For $n = 10, 20$ many more numbers of shuffles pass the test.
- However it disagrees that 6 is the cutoff point $k = 6$ for n close to 50.
- It was reproducible with code, so either changing the cards slightly has a big effect (deviation from normal?), or the code was wrong

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 7$	$k = 8$	$k = 9$	$k = 10$
$n = 10$	0	0	0	2.50E-08	0.097340543	0.143626961	0.731927124	0.006772995	0.092887862	0.466263746
$n = 20$	0	0	2.11E-11	0.003799097	0.281606428	0.363229759	0.647516686	0.278111308	0.373611047	0.457378459
$n = 30$	0	0	0	0	0	1.00E-12	4.93E-05	0.000941665	0.032542408	0.127411474
$n = 40$	0	0	0	0	1.11E-16	9.97E-08	0.00147718	0.011942079	0.097309442	0.244010238
$n = 50$	0	0	0	0	3.36E-12	1.08E-05	0.006830986	0.029446503	0.141784876	0.275091237
$n = 60$	0	0	0	0	4.04E-10	9.85E-05	0.016380498	0.057526003	0.197667004	0.305085452
$n = 70$	0	0	0	0	3.02E-09	0.0003118	0.024741265	0.085733568	0.241955138	0.325223196
$n = 80$	0	0	0	0	1.19E-09	0.000454637	0.03308458	0.102256466	0.268628667	0.353488198
$n = 90$	0	0	0	0	3.81E-11	0.000330876	0.032567333	0.111657155	0.276162625	0.376676234
$n = 100$	0	0	0	0	3.79E-14	7.68E-05	0.025397609	0.115023635	0.303789377	0.376385069

Table: A table of k , the estimated means of U_1 , test statistics and p values for the test on $k = 1, \dots, 10$ riffles.

Adding a cut

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- People sometimes add a cut inbetween shuffles, since riffling mixes up cards' neighbours, but cutting changes their positions
- Adding a cut seems to decrease the number of shuffles needed to get the same disorder by 1 except when measuring by U_1
- U_1 isn't affected by cycles

The normal model

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- Now want to optimise parameters
- μ seems to barely affect the disorder in the range [20, 32] which most human averages should be well inside
- distribution looks roughly symmetric with a peak at 26, however it is hard to tell - would need to use more samples
- σ seems to peak around 5 with a $\sigma = 0$ being roughly as good as $\sigma = 8$
- Adding a cut doesn't seem to change these values
- when we use these values, we still get a sharp cutoff at 6 shuffles

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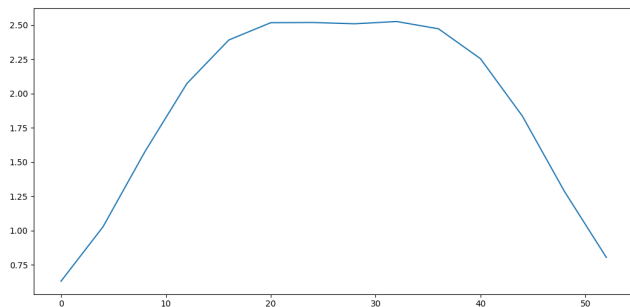


Figure: Average U_1 of 10000 decks shuffled once with $\mu = 0, 4, \dots, 52$

Experience of shuffler

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- The cutting mean makes almost no difference as seen above
- All shufflers have less than optimal standard deviation
- However, this doesn't affect our cutoff point of 6 shuffles
- This would suggest less experienced shufflers shuffle the cards better, if they riffle the same (which is unlikely)

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- To measure disorder, U_1 , r and s are the "best" measures
- It takes 6 shuffles to get to the point where the level of disorder in a sample (as measured by U_1) is indistinguishable from that of a perfectly random sample
- However there are still ways to tell that a sample isn't random (rising sequences)
- Adding a cut inbetween doesn't affect U_1 , but has roughly the same affect as shuffling one more time
- The way humans cut the cards doesn't affect this result
- Most humans cut with too little variance, but their mean doesn't matter

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Improvements to be made

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- Investigate other measures of disorder, particularly to do with real cards e.g. suits and the card game being played, or optimal strategies given known GSR probabilities
- Try a mixed measure of disorder
- Gather more data on packet riffing, in a controlled environment
- Test other distributions for cuts and riffles
- Could work out the exact probability of k shuffles passing using GSR model